

A Reproducible Post-Processing Framework for Multiple-Scattering Correction in Ultra-Small-Angle Neutron Scattering

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Synopsis

A modular post-processing framework is proposed for diagnosing and correcting multiple scattering in reduced USANS data while preserving reproducibility, metadata traceability, and user control.

Abstract

Ultra-small-angle neutron scattering (USANS) provides access to structural correlations on micrometre and sub-micrometre length scales, but quantitative interpretation can be affected by multiple scattering when samples are strongly scattering, thick, highly concentrated, or contain large characteristic structures. In such cases, the measured intensity can no longer be interpreted directly as a single-scattering profile, and apparent changes in low- Q slope, correlation length, particle size, or scattering invariant may arise from repeated scattering events rather than sample structure alone.

We present the design of a reproducible post-processing framework for multiple-scattering treatment in USANS data analysis. The framework is intended to operate after standard USANS reduction steps and before final model fitting, desmearing, or SANS-USANS stitching. Rather than replacing the core instrument reduction pipeline, the module provides an optional correction layer that preserves the original reduced data, records correction assumptions and parameters, and exports both corrected and uncorrected profiles for downstream analysis.

The proposed workflow separates diagnosis, correction, and reporting. Reduced one-dimensional intensity profiles, transmission information, sample thickness, uncertainty estimates, and relevant reduction metadata are used to assess whether multiple scattering is likely to be significant. When the assumptions of the selected correction model

are satisfied, the software applies an iterative convolution/deconvolution-based correction to estimate the corresponding single-scattering profile. All correction parameters, convergence metrics, and output data products are stored in a reproducible form.

This work establishes the software design and methodological basis for incorporating multiple-scattering correction into routine USANS analysis. Experimental validation, benchmark examples, and integration with production reduction workflows will be reported as the implementation matures.

Keywords: USANS; multiple scattering; neutron scattering; data reduction; post-processing; reproducibility; software framework

1 Introduction

Ultra-small-angle neutron scattering (USANS) is a powerful technique for characterizing structural features on micrometre and sub-micrometre length scales. By extending the accessible momentum-transfer range below that of conventional small-angle neutron scattering (SANS), USANS enables the study of large pores, aggregates, precipitates, phase-separated domains, and hierarchical mesoscale structures in soft matter, complex fluids, porous media, geological materials, and engineering systems. The USANS instrument at the Spallation Neutron Source (SNS) at Oak Ridge National Laboratory (ORNL) is based on a Bonse–Hart double-crystal diffractometer geometry (Bonse & Hart, 1966; Agamalian *et al.*, 1997; Heller *et al.*, 2018). Its time-of-flight implementation takes advantage of the pulsed neutron source to separate wavelength orders in time, allowing multiple harmonic banks to be treated independently or combined during data reduction (Oak Ridge National Laboratory, 2021).

Reliable USANS interpretation depends on accurate and reproducible data reduction. As in SANS, raw measured intensities must be corrected for instrumental and sample-dependent effects before they can be interpreted as scattering cross sections. Common reduction operations include monitor normalization, conversion from analyzer angle to momentum transfer, stitching of overlapping scans, intensity rescaling, transmission correction, empty-scan or background subtraction, uncertainty propagation, and optional binning. Existing neutron-scattering software frameworks, including Mantid, SasView, the NIST SANS/USANS reduction tools, and the ORNL *drtans* framework, demonstrate the importance of transparent, instrument-aware, and reproducible reduction workflows for user-facility science (Arnold *et al.*, 2014; Doucet *et al.*, 2017; Heller *et al.*, 2022).

For USANS, the reduction problem is especially sensitive to very small scattering angles, high dynamic range, instrumental resolution, slit smearing, transmission, and sample multiple scattering. The scientific motivation for USANS measurements often involves strongly scattering samples with large characteristic length scales. These same conditions can increase the probability that a neutron is scattered more than once before reaching the detector. When this occurs, the measured intensity

is no longer a direct representation of the desired single-scattering profile. Multiple scattering can redistribute intensity in reciprocal space and distort apparent low- Q features, potentially affecting derived quantities such as correlation length, particle size, fractal exponent, Porod behavior, or scattering invariant (?; ?; ?).

In the single-scattering approximation, the reduced intensity is interpreted as being proportional to the sample scattering cross section after correction for instrumental effects. Multiple scattering breaks this relationship because the detected neutron may have undergone a sequence of scattering events. In an independent-event description, the measured profile can be represented conceptually as a weighted sum of repeated self-convolutions of the single-scattering distribution,

$$I_{\text{meas}}(Q) \simeq \sum_{n=1}^{\infty} w_n I^{(*n)}(Q), \quad (1)$$

where $I^{(*n)}(Q)$ denotes the n -fold convolution of the single-scattering profile and w_n is the probability weight for n scattering events. Although practical correction methods require additional assumptions about sample transmission, thickness, resolution, and normalization, this expression illustrates the central software challenge: the measured reduced curve may contain contributions from repeated scattering events and therefore cannot always be used directly for quantitative modeling.

Several theoretical and computational approaches have been developed for treating multiple scattering in small-angle scattering and USANS, including analytical independent-scattering models, Fourier-transform methods, iterative deconvolution, and Monte Carlo simulations (?; ?; ?). These studies show that multiple scattering is not merely an experimental artifact, but a predictable consequence of sample thickness, scattering strength, wavelength, concentration, and characteristic structural length scale. In practice, users may reduce multiple scattering experimentally by decreasing sample thickness, contrast, or concentration, but these choices are not always compatible with available sample environments, beam time, signal-to-noise requirements, or the scientific objective of measuring weak low- Q features.

Despite the importance of the effect, multiple-scattering correction is not always available as a routine, transparent, and reproducible part of USANS data analysis. Users may recognize possible multiple scattering from sample transmission, thickness dependence, or unexpected changes in curve shape, but the subsequent correction is often performed outside the main reduction workflow using ad hoc scripts or manual post-processing. This separation makes it difficult to record correction assumptions, preserve intermediate data products, propagate uncertainty consistently, or compare corrected and uncorrected profiles in a traceable way.

Here we present the design of a multiple-scattering post-processing framework for USANS data. The goal is not to introduce a new scattering theory or to replace the core instrument reduction pipeline. Instead, the framework provides a reproducible software layer that operates on reduced

one-dimensional USANS intensity profiles and associated metadata. The module is designed to support three practical tasks: diagnosing whether multiple scattering is likely to be significant, applying a documented correction method when its assumptions are appropriate, and exporting corrected data together with the metadata required to reproduce and evaluate the correction.

The proposed workflow is intentionally separated from standard reduction operations such as monitor normalization, scan stitching, transmission correction, and empty-scan subtraction. This separation allows the original reduced data to remain available, while enabling users to apply multiple-scattering correction as an optional post-processing step before model fitting, desmearing, or SANS–USANS stitching. The software records the selected correction model, sample parameters, convergence behavior, and generated output files, allowing corrected and uncorrected results to be compared directly. This design is consistent with the broader goal of user-facility software: to provide robust and reproducible correction tools while keeping scientific assumptions and interpretation visible to the user.

The remainder of this paper is organized as follows. Section ?? summarizes the multiple-scattering problem in the context of USANS measurements. Section 3 describes the proposed post-processing workflow and its relationship to standard USANS reduction. Section 4 presents the software architecture, data model, configuration system, and output products. Section 5 outlines the planned validation strategy using simulated and experimental datasets. Finally, Section 6 discusses limitations, expected use cases, and future integration with production USANS reduction workflows.

2 Multiple-Scattering Methodology

The multiple-scattering correction implemented in this work follows the algebraic convolution formulation developed by Tung *et al.* (Tung *et al.*, 2026). The method treats multiple scattering as a sequence of statistically independent scattering events in an isotropic transverse-momentum plane. Under this approximation, the number of scattering events experienced by a neutron is modeled as a Poisson random variable. If n is the number of scattering events and μ is the mean event count, then

$$P_n = e^{-\mu} \frac{\mu^n}{n!}. \quad (2)$$

The transmission associated with the small-angle scattering process is then

$$T = P_0 = e^{-\mu}, \quad (3)$$

so that

$$\mu = -\ln T. \quad (4)$$

In practice, the experimentally measured transmission may contain attenuation from absorption, incoherent scattering, and other processes in addition to small-angle scattering. Therefore, the transmission used in the correction must be interpreted consistently with the assumptions of the

131 chosen correction model.

132 Let $\hat{I}(Q)$ denote the primary single-scattering intensity. The method normalizes this profile with
 133 respect to the natural isotropic measure of the two-dimensional transverse momentum plane,

$$\int_0^\infty 2\pi Q \hat{I}(Q) dQ = I_0, \quad (5)$$

134 where I_0 is the integrated single-scattering intensity scale. The normalized primary profile,

$$J(Q) = \frac{\hat{I}(Q)}{I_0}, \quad (6)$$

135 can then be interpreted as the probability density for a single scattering event in transverse mo-
 136 mentum space.

137 For two isotropic scattering profiles $f(Q)$ and $g(Q)$, convolution in the transverse momentum plane
 138 is defined as

$$(f * g)(Q) = \int_0^\infty q dq \int_0^{2\pi} d\Phi f(q)g\left(\sqrt{Q^2 + q^2 - 2Qq \cos \Phi}\right), \quad (7)$$

139 where Φ is the relative azimuthal angle. The n -fold self-convolution of the primary profile, denoted
 140 $J^{(*n)}(Q)$, represents the distribution of neutrons that have scattered exactly n times.

141 The apparent intensity measured after multiple scattering can therefore be written as a Poisson-
 142 weighted sum of self-convolutions,

$$I^{\text{App}}(Q) = P_0 I_0 \delta(Q) + I_0 \sum_{n=1}^\infty P_n J^{(*n)}(Q), \quad (8)$$

143 where the first term represents the unscattered direct-beam contribution. For reduced SANS or
 144 USANS profiles, this direct-beam term is normally removed or treated separately, and the correction
 145 focuses on the scattered component.

146 The central feature of the algebraic method is to avoid direct numerical convolution or explicit
 147 Fourier/Hankel transformation of truncated experimental data. Instead, the normalized primary
 148 profile is expanded in an orthonormal basis $\{u_\ell(Q)\}$ adapted to the isotropic measure $2\pi Q dQ$,

$$J(Q) = \sum_{\ell=1}^L \alpha_\ell^{(1)} u_\ell(Q), \quad (9)$$

149 with orthonormality condition

$$\langle u_\ell, u_m \rangle = \int_0^\infty 2\pi Q u_\ell(Q) u_m(Q) dQ = \delta_{\ell m}. \quad (10)$$

150 The convolution of two basis functions is represented by precomputed structure constants,

$$(u_j * u_k)(Q) = \sum_{\ell=1}^L C_{jk}^{\ell} u_{\ell}(Q). \quad (11)$$

151 As a result, convolution of arbitrary profiles becomes a finite-dimensional tensor contraction. The
152 coefficients of the n -fold self-convolution are obtained recursively as

$$\alpha_{\ell}^{(n)} = \sum_{j,k} C_{jk}^{\ell} \alpha_j^{(1)} \alpha_k^{(n-1)}, \quad n \geq 2. \quad (12)$$

153 For example, the double-scattering contribution is

$$\alpha_{\ell}^{(2)} = \sum_{j,k} C_{jk}^{\ell} \alpha_j^{(1)} \alpha_k^{(1)}, \quad (13)$$

154 and the triple-scattering contribution is obtained by convolving the primary coefficients with the
155 double-scattering coefficients,

$$\alpha_{\ell}^{(3)} = \sum_{j,k} C_{jk}^{\ell} \alpha_j^{(1)} \alpha_k^{(2)}. \quad (14)$$

156 The apparent multiple-scattered profile can then be reconstructed in the same basis,

$$I^{\text{App}}(Q) = I_0 \sum_{\ell=1}^L \beta_{\ell} u_{\ell}(Q), \quad (15)$$

157 where the apparent coefficients are given by the Poisson-weighted sum

$$\beta_{\ell} = \sum_{n=1}^N P_n \alpha_{\ell}^{(n)}. \quad (16)$$

158 Here, N is a finite truncation order. Because P_n decays rapidly with n , the series can be truncated
159 once higher-order scattering probabilities become negligible.

160 The inverse problem is formulated as the recovery of the primary coefficients $\{\alpha_{\ell}^{(1)}\}$ from the
161 observed apparent intensity. Given measured data $I^{\text{Obs}}(Q_i)$ with uncertainties ΔI_i , the correction
162 is obtained by minimizing the weighted residual between the observed apparent spectrum and the
163 forward multiple-scattering model,

$$\chi^2 = \sum_i \frac{[I^{\text{Obs}}(Q_i) - I^{\text{App}}(Q_i; \{\alpha_{\ell}^{(1)}\})]^2}{\Delta I_i^2}. \quad (17)$$

164 The corrected single-scattering intensity is then reconstructed from the optimized first-order coef-

ficients,

$$\hat{I}^{\text{Fit}}(Q) = I_0 \sum_{\ell=1}^L \alpha_{\ell}^{(1)} u_{\ell}(Q). \quad (18)$$

This formulation is well suited for software implementation because the computationally expensive parts of the method can be separated from the data-dependent fitting. The basis functions and convolution tensor C_{jk}^{ℓ} can be constructed or cached for a chosen basis and truncation order, while each experimental dataset only requires projection, forward evaluation of the Poisson-weighted convolution series, and optimization of the primary coefficients. Compared with transform-based correction methods, the algebraic representation avoids direct integral transforms of finite experimental Q windows and reduces reliance on model-dependent extrapolation outside the measured range. For USANS data, this is particularly useful because low- Q coverage, slit smearing, transmission effects, and SANS–USANS stitching can make extrapolation-based correction unstable or difficult to reproduce.

The physical origin of multiple scattering and the corresponding mathematical formulation are summarized schematically in Fig. 1. As illustrated in Fig. 1(a), multiple scattering arises when a neutron undergoes two or more successive scattering events before reaching the detector, leading to broadening and redistribution of the measured angular intensity profile relative to the single-scattering case. Fig. 1(b) summarizes the forward description used in the present work: the primary single-scattering intensity $\hat{I}(Q)$ is normalized to $J(Q)$, propagated through repeated self-convolutions, and combined through Poisson weights determined by the transmission. The same panel also highlights the algebraic basis-expansion representation used in the software implementation, in which convolution is evaluated through precomputed structure constants and higher-order scattering contributions are generated recursively.

In the software framework proposed here, this methodology is implemented as an optional post-processing layer applied after standard USANS reduction. The original reduced profile is preserved, while the corrected primary profile, optimized basis coefficients, transmission value, truncation order, convergence metrics, and correction configuration are exported as separate data products. This design allows users to compare corrected and uncorrected results directly and to evaluate whether the assumptions of the algebraic multiple-scattering correction are appropriate for a given dataset.

3 Post-Processing Workflow

The multiple-scattering correction is designed as an optional post-processing step applied after standard USANS data reduction. It does not replace core reduction operations such as monitor normalization, scan stitching, transmission correction, empty-scan subtraction, or uncertainty propagation. Instead, it operates on the reduced one-dimensional intensity profile, $I(Q)$, together with the associated uncertainties and metadata, and produces an additional corrected estimate

of the primary single-scattering profile. The original reduced data are preserved throughout the workflow.

The required inputs include the reduced intensity profile, uncertainty estimates, sample transmission, and reduction metadata. When available, sample thickness, wavelength-bank information, resolution description, empty-scan processing history, and absolute scaling factors are also recorded and used for diagnostic reporting. These quantities allow the software to evaluate whether multiple scattering is likely to be important and whether the assumptions of the selected correction model are appropriate for the dataset.

The workflow consists conceptually of three stages: diagnosis, correction, and reporting. In the diagnostic stage, the software summarizes indicators of possible multiple scattering, such as low transmission, strong low- Q intensity, thickness dependence, or user-provided sample parameters. This step is not intended to make the scientific decision automatically, but to provide reproducible information that helps the user decide whether correction is justified. If correction is requested, the software applies the algebraic convolution method described in Section 2 to estimate the corresponding primary single-scattering intensity. The correction is carried out as a forward-inverse problem: a candidate primary profile is propagated through the Poisson-weighted multiple-scattering model and optimized until the modeled apparent intensity agrees with the observed reduced data within the specified uncertainty model.

The outputs include the corrected intensity profile, the original uncorrected profile, propagated or estimated uncertainties, correction parameters, diagnostic metrics, convergence information, and links to the input reduction files. By writing corrected and uncorrected results as separate data products, the framework avoids overwriting or obscuring the standard reduction output. This non-destructive design allows users to compare the effect of multiple-scattering correction directly and to report the correction assumptions in a reproducible form.

The overall post-processing procedure is summarized schematically in Fig. 2. The workflow begins with standard USANS reduction, which generates the reduced one-dimensional intensity profile together with the associated uncertainties, transmission value, and reduction metadata. These outputs form the input to the multiple-scattering post-processing stage. A diagnostic summary is first generated to provide reproducible indicators of whether multiple scattering is likely to be significant. If correction is requested and considered justified, the reduced profile is processed using the algebraic multiple-scattering method described in Section 2. Otherwise, the reduced data are reported without modification. In either case, the workflow preserves the original reduced result and records the relevant diagnostic and processing information together with the final output products.

4 Software Design and Implementation

The multiple-scattering correction module is designed as a post-processing layer that can be attached to the reduced-data output of the USANS reduction framework. The implementation separates data representation, correction configuration, convolution-model construction, numerical optimization, and output reporting. This separation is intended to make the correction reproducible and extensible while keeping the original reduced data unchanged.

At the data level, the central object is a reduced scattering profile, represented conceptually by a **ReducedProfile** class. This object contains the momentum-transfer array Q , the reduced intensity $I(Q)$, the uncertainty $\Delta I(Q)$, and a metadata dictionary describing the origin of the data. The metadata include the input file path, reduction configuration, sample name, transmission value, sample thickness when available, wavelength-bank or harmonic information, empty-scan processing history, and any absolute-scaling or resolution information available from the reduction stage. The multiple-scattering correction does not modify this object in place. Instead, the corrected result is stored in a separate **CorrectedProfile** object that contains the reconstructed primary intensity, corrected uncertainty estimate, correction status, and a reference back to the original **ReducedProfile**. This design allows corrected and uncorrected data to be compared directly and prevents the correction from overwriting the standard reduction output.

Correction parameters are specified through a structured configuration object, here referred to as **MultipleScatteringConfig**. This configuration records whether correction is enabled, which correction model is selected, the basis size L , the maximum scattering order N , the basis length-scale parameter a , convergence tolerances, iteration limits, fitting bounds or regularization options, and output preferences. It also records how the transmission value is obtained, for example whether it is read from the reduction metadata, supplied manually, or estimated from a separate measurement. Keeping these parameters in a configuration object makes the correction reproducible and allows the same workflow to be applied consistently across multiple samples.

The algebraic convolution method is implemented through a model object, denoted here as **AlgebraicConvolution**. This class is responsible for constructing the orthonormal basis functions $u_\ell(Q)$, evaluating or loading their Hankel transforms when required, and building the convolution tensor C_{jk}^ℓ . Since the basis and structure constants depend only on the chosen basis family, truncation size, and numerical grid, these quantities can be cached and reused across datasets. The model object also provides the core numerical operations used by the correction:

- **expand_profile**: project or fit a measured profile into basis coefficients;
- **convolve_coefficients**: compute the coefficients of a self-convolution using C_{jk}^ℓ ;
- **generate_orders**: recursively compute $\alpha_\ell^{(n)}$ for $n = 1, \dots, N$;
- **forward**: generate the apparent multiple-scattered intensity from a candidate primary profile;

- **recover_primary**: optimize the primary coefficients against the observed reduced profile.

These functions map directly onto the mathematical formulation in Section 2: repeated scattering is evaluated through recursive tensor contractions, and the apparent intensity is assembled from the Poisson-weighted sum of the resulting self-convolution orders.

The execution of a correction is coordinated by a higher-level **MultipleScatteringCorrector** class. This class takes a **ReducedProfile** and a **MultipleScatteringConfig** as input, validates the required metadata, constructs the corresponding **AlgebraicConvolutionModel**, and runs the diagnostic and correction workflow. The diagnostic stage checks whether the inputs are physically and numerically suitable for correction, including the availability and range of transmission, the number of valid Q points, the presence of non-positive intensities or large uncertainties, and optional indicators such as sample thickness or unusually strong low- Q scattering. These diagnostics are reported to the user but do not automatically replace scientific judgement.

When correction is requested, the corrector solves a forward-inverse problem. A candidate primary profile is represented by its basis coefficients $\{\alpha_\ell^{(1)}\}$. These coefficients are propagated through the recursive convolution model to generate higher-order scattering contributions, which are then combined using the Poisson weights determined from the transmission. The resulting apparent intensity is compared with the observed reduced profile using a weighted least-squares objective,

$$\chi^2 = \sum_i \frac{[I^{\text{Obs}}(Q_i) - I^{\text{App}}(Q_i; \{\alpha_\ell^{(1)}\})]^2}{\Delta I_i^2}. \quad (19)$$

The optimized coefficients define the reconstructed primary single-scattering profile. The correction result includes the fitted apparent intensity, the recovered primary intensity, residuals, convergence status, fitted coefficients, and warnings generated during validation or optimization.

The module writes all outputs non-destructively. A typical output set includes the original reduced profile, the corrected primary profile, the fitted apparent profile, uncertainty estimates, the optimized coefficient vector, diagnostic metrics, convergence information, and a machine-readable summary of the correction configuration. In addition, a human-readable report is generated to summarize the correction in a form suitable for review and publication. This report includes the input data provenance, transmission value, basis parameters, truncation order, optimizer settings, convergence behavior, warnings, and output file names. Together, these outputs make the correction traceable and reproducible while preserving the user's ability to inspect the uncorrected reduced data.

The proposed software architecture is summarized schematically in Fig. 3. The implementation is organized around a small number of central objects. A **ReducedProfile** object stores the reduced scattering data and associated metadata, while a **MultipleScatteringConfig** object records the correction settings. These two objects are passed to the **MultipleScatteringCorrector**, which serves as the main controller of the workflow. The corrector interacts with the **AlgebraicConvolutionModel**,

which encapsulates the basis-expansion representation, convolution tensor, forward multiple-scattering calculation, and inverse recovery of the primary profile. After validation, diagnostics, and correction, the results are stored in a `CorrectedProfile` object and passed to the reporting/output layer, which writes the corrected and uncorrected profiles together with diagnostic metrics, convergence information, and configuration metadata. This architecture keeps the numerical model, workflow control, and input/output responsibilities clearly separated while preserving the original reduced data non-destructively.

5 Validation Plan and Example Applications

The validation of the multiple-scattering post-processing framework will be performed using both simulated and experimental datasets. Simulated data will be used to generate known single-scattering profiles and controlled multiple-scattering distortions, allowing the correction accuracy to be evaluated quantitatively. Experimental validation will use USANS measurements collected under conditions where multiple scattering is expected to vary systematically, for example through changes in sample thickness, concentration, or transmission.

Representative validation metrics will include recovery of the input single-scattering profile, stability of fitted structural parameters, sensitivity to noise, convergence behavior, and comparison between corrected and uncorrected results. The validation examples will also test the reproducibility of configuration-driven execution and the completeness of exported metadata.

6 Discussion

The proposed framework addresses a practical gap between established multiple-scattering theory and routine USANS data analysis. By implementing correction as a reproducible post-processing layer, the software allows users to evaluate multiple-scattering effects without altering the core reduction pipeline or overwriting the original reduced data. This approach is particularly suitable for user-facility workflows, where data provenance, transparency, and optional correction steps are essential.

The framework is not intended to remove the need for careful experimental design. Sample thickness, concentration, contrast, and transmission should still be considered during measurement planning. However, when multiple scattering cannot be avoided experimentally, a documented post-processing correction provides a practical way to assess its impact and estimate a corrected single-scattering profile.

The initial implementation will focus on correction methods appropriate for independent multiple scattering. More complex effects, including coherent multiple scattering, full resolution coupling, and joint desmearing/multiple-scattering correction, may require additional development and validation. These topics are treated as future extensions rather than assumptions of the first imple-

mentation.

7 Conclusions

A reproducible post-processing framework for multiple-scattering correction in USANS data analysis has been proposed. The framework is designed to operate after standard data reduction and before final model fitting, desmearing, or SANS-USANS stitching. By preserving the original reduced data and exporting correction metadata, the software supports transparent comparison between corrected and uncorrected profiles.

The planned implementation separates diagnosis, correction, and reporting, allowing multiple-scattering effects to be evaluated systematically. The modular design provides a foundation for future correction models, validation datasets, and integration with production USANS reduction workflows. Once implemented and validated, the framework will provide USANS users with a practical and reproducible tool for treating multiple scattering in strongly scattering samples.

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Conflicts of interest: The authors declare no conflicts of interest.

Data availability: The data supporting the findings of this study are available within the article and from the USANSRed repository at <https://github.com/neutrons/usansred>.

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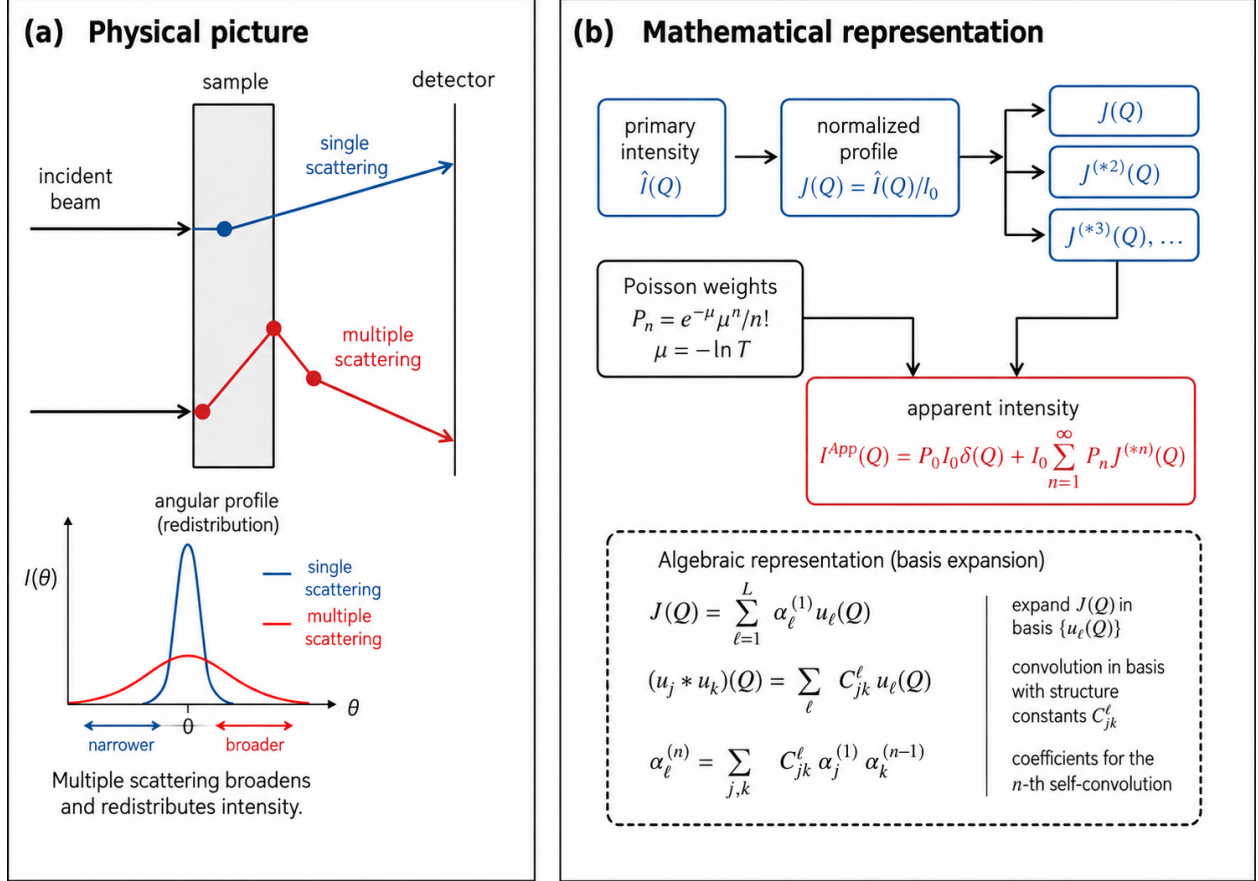


Figure 1: Schematic illustration of multiple scattering and its mathematical representation. (a) Physical picture of single and multiple scattering in a sample. In the single-scattering approximation, a neutron is scattered at most once before reaching the detector. In multiple scattering, a neutron undergoes two or more successive scattering events before exiting the sample, which broadens and redistributes the measured angular intensity profile. (b) Mathematical representation used in the present work. The primary single-scattering intensity $\hat{I}(Q)$ is normalized to $J(Q) = \hat{I}(Q)/I_0$, and repeated self-convolutions $J(Q)$, $J^{(*2)}(Q)$, $J^{(*3)}(Q)$, ... are combined using Poisson weights $P_n = e^{-\mu} \mu^n / n!$, where $\mu = -\ln T$, to produce the apparent multiple-scattered intensity $I^{APP}(Q)$. In the algebraic formulation, the normalized primary profile is expanded in an orthonormal basis, the convolution is represented through structure constants C_{jk}^{ℓ} , and the coefficients of higher-order self-convolutions are generated recursively.

Post-processing workflow for multiple-scattering correction

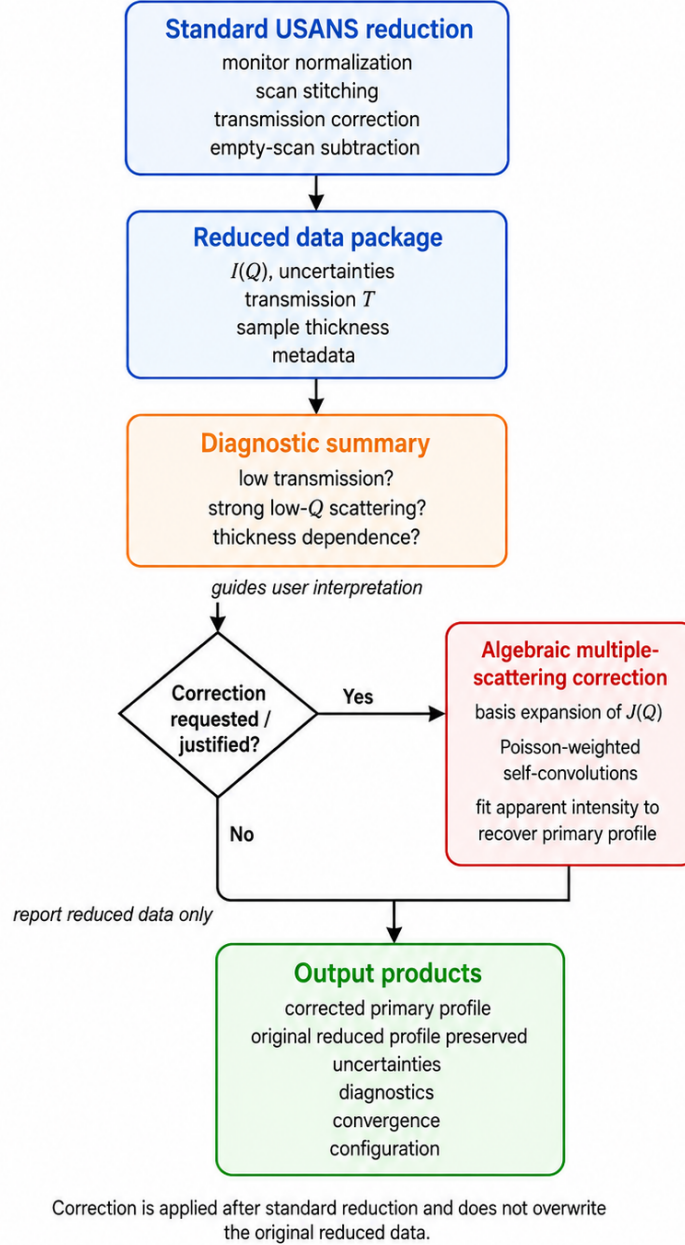
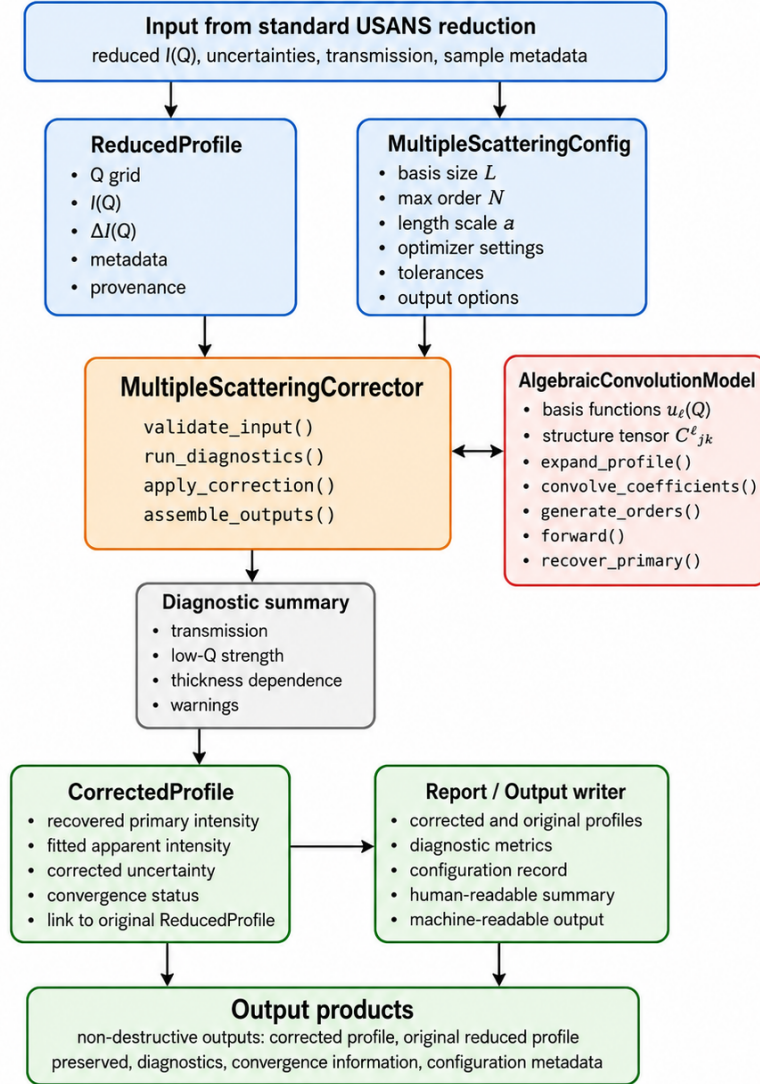


Figure 2: Post-processing workflow for multiple-scattering correction. Standard USANS reduction first produces a reduced data package containing the one-dimensional intensity profile, associated uncertainties, transmission information, and metadata. A diagnostic summary is then generated to assess whether multiple scattering is likely to be significant, based on indicators such as transmission, low- Q scattering strength, and thickness dependence. If correction is requested and considered justified, the reduced profile is processed using the algebraic multiple-scattering correction, in which the primary profile is represented in a basis-expansion form and propagated through Poisson-weighted self-convolutions to reproduce the apparent intensity. The output includes the corrected primary profile together with the original reduced profile, uncertainties, diagnostic information, convergence metrics, and configuration metadata. If correction is not applied, the workflow reports the reduced data without modification.

Software architecture of the multiple-scattering correction module



Correction is applied non-destructively; the original reduced data are preserved.

Figure 3: Software architecture of the multiple-scattering correction module. Reduced USANS data and correction settings are represented by the `ReducedProfile` and `MultipleScatteringConfig` objects, respectively. These are passed to the `MultipleScatteringCorrector`, which coordinates input validation, diagnostic evaluation, correction execution, and output assembly. The numerical multiple-scattering model is encapsulated in the `AlgebraicConvolutionModel`, which provides the basis functions, convolution tensor, forward model, and primary-profile recovery routines. The correction produces a `CorrectedProfile` object linked to the original reduced data, together with diagnostic summaries and report/output writers that generate both human-readable and machine-readable output products. The design is non-destructive: the corrected results are written separately and the original reduced data are preserved.